Apex Global Bank

Quantitative Research & Treasury

Bank Balance Sheet Optimization

How Banks Make Money, Manage Risk, and Navigate
a Maze of Regulations — Explained for Everyone

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This document tells the story of how a major bank optimizes its balance sheet: what that means, why it matters, how the mathematics works, and how it generates billions of dollars in annual revenue. It is written to be accessible to anyone who has ever balanced a personal budget or wondered how banks actually make money — while retaining full technical rigour for the quants and risk managers who will use these models in production. Sections 3–8 contain material non-public information. Distribution restricted to authorized personnel.

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1 Introduction: The World's Most Complex Household Budget

1.1 What Is a Balance Sheet, and Why Does It Need Optimizing?

Every business has a balance sheet — a snapshot that answers two questions: *What do you own?* and *Where did the money come from to buy it?* Your own personal balance sheet might look like this: you own a house ($400K), a car ($20K), and savings ($30K). You owe a mortgage ($300K) and a car loan ($15K). The difference — what you own minus what you owe — is your equity ($135K).

A bank's balance sheet works on exactly the same principle, just at a scale that strains the imagination. Apex Global Bank's balance sheet stands at $285 billion. JPMorgan's is $3.9 trillion — larger than the GDP of Germany.

Analogy

Think of a bank as a giant financial intermediary. It borrows money cheaply (from your savings deposits, paying you 1–2%) and lends it out expensively (mortgages at 6%, corporate loans at 7%). The spread between those rates — the "net interest margin" — is the bank's core business. Balance sheet optimization is the discipline of making that spread as large and as safe as possible.

On the **asset side** of the balance sheet, the bank holds loans, bonds, cash, and derivatives. On the **liability side**, it holds deposits, borrowed money, and issued debt. The difference between assets and liabilities is the bank's equity — the "skin in the game" that shareholders have put in.

Balance sheet optimization is the discipline of constructing and managing this portfolio to maximize risk-adjusted return subject to regulatory constraints, liquidity requirements, and the bank's own risk appetite. It sounds simple in one sentence. In practice, it is one of the most complex multi-dimensional optimization problems in applied mathematics.

At the highest level, the problem is:

$$\max_{\mathbf{a},\mathbf{l}} \quad \mathrm{RORWA}(\mathbf{a},\mathbf{l}) \quad \text{subject to} \quad \mathbf{g}(\mathbf{a},\mathbf{l}) \leq \mathbf{0} \tag{1}$$

In plain terms

Maximize return on capital (RORWA) by choosing the best mix of assets ($\mathbf{a}$) and liabilities ($\mathbf{l}$) — but stay inside the rules ($\mathbf{g} \leq \mathbf{0}$ encodes dozens of regulatory constraints).

Unlike trading book optimization, which operates on a horizon of hours to days, balance sheet optimization operates on horizons of months to years. It is a strategic planning function that touches every product line: commercial lending, retail deposits, fixed-income securities portfolios, derivatives books, repo/securities finance, and off-balance-sheet vehicles.

1.2 Why It Matters: Billions of Dollars Are on the Table

Here is a number that puts everything into perspective: at JPMorgan, a 5 basis point (0.05%) improvement in return on assets is worth approximately $2 billion per year. That is the magnitude of the prize for getting balance sheet optimization right.

The levers available include:

- **Capital efficiency**: Hold the same loans but use less shareholder capital by choosing assets with lower regulatory "risk weights". The freed capital can be redeployed or returned to shareholders via buybacks.
- **Funding cost reduction**: Borrow at the lowest possible cost consistent with stability. In 2023, banks that locked in long-term funding cheaply before the rate hike cycle saved hundreds of basis points relative to peers who relied on short-term floating-rate paper.
- **NII maximization**: Position the bank's interest rate sensitivity to benefit from expected rate moves. A bank with the right duration gap captured an extra $6.8 billion in net interest income when rates rose 200bps in 2022–2023.
- **Smart regulatory navigation**: Choosing product structures that achieve the same economic goal with lower capital consumption. Example: a guarantee costs far less capital than a direct loan to the same counterparty.
- **Liquidity buffer efficiency**: Minimizing the drag from mandatory "rainy-day fund" assets while staying compliant.

1.3 The Regulatory Maze

After the 2008 financial crisis, regulators worldwide decided that banks had been far too thinly capitalised and far too reliant on short-term, flighty funding. They built a dense framework of rules to prevent a repeat. Every optimization decision at Apex Global Bank must fit inside this framework:

Table 1: Key Regulatory Constraints on the Bank Balance Sheet

Regulation	What It Measures	Minimum	Apex Today
Basel III / 12 CFR 3	Core equity vs. risk	4.5% (+ 2.5% buffer)	13.0%
Basel III / 12 CFR 3	Broader capital vs. risk	6.0% (+ buffer)	14.5%
Basel III / 12 CFR 3	Total capital vs. risk	8.0% (+ buffer)	16.2%
Basel III / 12 CFR 3	Capital vs. all assets	3.0% (SLR)	7.8%
Basel III / 12 CFR 249	30-day liquidity buffer	100%	118%
Basel III / 12 CFR 249	1-year funding stability	100%	112%
Dodd-Frank / 12 CFR 252	Worst recession CET1 floor	4.5% regulatory	8.7% trough
Basel IV / FRTB	Trading book capital floor	Binding from 2025	—
Concentration rules	Single-name exposure cap	25% of CET1	Monitored

Each of these constraints carves away part of the feasible solution space. The art of balance sheet optimization is finding the highest-return operating point within what remains.

2 The Optimization Problem: Putting It All Into One Equation

2.1 Setting Up the Problem

To write this problem mathematically, we first need to define what we are choosing. Imagine the bank's balance sheet as a giant spreadsheet that updates every quarter. The rows are asset classes and liability classes. The optimization asks: by how much should we grow or shrink each row?

Let $t = 0, 1, \ldots, T$ index quarterly periods. The state of the balance sheet at time t is described by:

- $A_k(t)$: balance in asset class k (loans, AFS securities, HTM securities, derivatives, cash, other), for $k = 1, \ldots, K_A$
- $L_j(t)$: balance in liability class j (demand deposits, time deposits, FHLB advances, senior unsecured debt, repo, equity), for $j = 1, \ldots, K_L$
- $r_k^A(t)$: yield on asset class k at time t
- $r_j^L(t)$: cost of liability class j at time t
- $\rho_k(t)$: risk weight for asset class k (how much capital it consumes under the regulatory rulebook)
- q_t: macroeconomic state vector (GDP growth, unemployment, rates)

The decision variables are the *incremental* changes at each step: $\Delta A_k(t)$ and $\Delta L_j(t)$ for all asset/liability classes and time periods.

The first hard constraint is the most fundamental fact about a bank:

$$\sum_k A_k(t) = \sum_j L_j(t), \qquad \forall\, t \tag{2}$$

In plain terms

You can only lend out what you have borrowed (plus your equity). Total assets always equal total liabilities plus equity. This is the accounting identity that ties the whole system together.

2.2 What We Are Trying to Maximize

The objective is to maximize the net present value of risk-adjusted earnings over the planning horizon:

$$\max_{\{\Delta A_k(t),\, \Delta L_j(t)\}} \sum_{t=0}^{T} \delta^t \left[\mathrm{NII}(t) + \mathrm{Fee}(t) - \mathrm{Credit_Loss}(t) - \mathrm{OpRisk}(t) - \mathrm{FTP_Cost}(t)\right] \tag{3}$$

where $\delta = 1/(1+h)$ is the discount factor, h is the bank's cost of equity (what shareholders expect as a minimum return), and the terms inside the brackets are:

- **NII**: Net Interest Income — interest earned on loans and bonds, minus interest paid on deposits and borrowed money. This is the core revenue engine.

- **Fee**: Non-interest income — trading revenues, advisory fees, transaction fees.
- **Credit_Loss**: Loans that go bad. In a recession, this number can explode.
- **OpRisk**: Operational risk losses — fines, fraud, systems failures.
- **FTP_Cost**: The internal cost of funding allocated to each business line (explained in detail in Section 6).

Management prefers to think in terms of RORWA — Return on Risk-Weighted Assets — because it accounts for how much capital each activity consumes:

$$\mathrm{RORWA}(t) = \frac{\mathrm{Net_Income_after_tax}(t)}{\mathrm{RWA}(t)} \tag{4}$$

Analogy

Think of RORWA as the efficiency score of the whole bank. A bank that earns \$10B on \$200B of risk-weighted assets (5% RORWA) is far more efficient than one that earns \$10B on \$500B of risk-weighted assets (2% RORWA), even if headline profits look the same.

2.3 Net Interest Income: The Engine Room

$$\mathrm{NII}(t) = \sum_k r_k^A(t) \cdot A_k(t) - \sum_j r_j^L(t) \cdot L_j(t) \tag{5}$$

In plain terms

NII is simply: (interest rate on each asset × balance) summed up, minus (interest rate on each liability × balance) summed up. If the bank lends \$100B at 6% and funds that with deposits at 1%, NII is \$5B. Optimization is about widening that gap — or at least protecting it.

The net interest margin (NIM) expresses NII as a percentage of earning assets — a benchmark for comparing banks of different sizes:

$$\mathrm{NIM}(t) = \frac{\mathrm{NII}(t)}{\mathrm{Average_Earning_Assets}(t)} \tag{6}$$

2.4 The Capital Constraints: How Much of Your Own Money?

The most fundamental rule in post-2008 banking is that banks must fund themselves with enough of their own equity to absorb unexpected losses. The CET1 (Common Equity Tier 1) ratio is the headline measure:

$$\frac{\mathrm{CET1}(t)}{\mathrm{RWA}(t)} \geq \kappa_{\mathrm{CET1}}, \qquad \forall\, t \tag{7}$$

where $\kappa_{\mathrm{CET1}} = 7\%$ in normal conditions (4.5% minimum + 2.5% Conservation Buffer) and 6.5% under a severe stress scenario.

Risk-weighted assets (RWA) are the denominator — not the raw balance of every asset, but each asset scaled by how risky it is:

$$\mathrm{RWA}(t) = \sum_k \rho_k(t) \cdot A_k(t) \tag{8}$$

Analogy

A mortgage on a house in New York backed by 20% equity gets a risk weight of roughly 50%, while a loan to a junk-rated company might get 150%. Regulators invented risk weights so that banks would not just pile into the riskiest loans because they yield the most. The trick is to find assets that yield well *relative* to their risk weight.

The leverage ratio is a backstop — it ignores risk weights entirely and simply asks: is there enough Tier 1 capital against all assets?

$$\frac{\mathrm{Tier1}(t)}{\mathrm{Total_Leverage_Exposure}(t)} \geq \kappa_{\mathrm{SLR}} \tag{9}$$

Real World

The leverage ratio is specifically there to prevent banks from gaming risk weights. Before 2008, some European banks ran leverage ratios of 40:1 — for every \$1 of equity, they held \$40 of assets. A 2.5% loss on assets wiped them out. Today the minimum leverage ratio is 3%—meaning the maximum leverage is roughly 33:1. Apex runs at 7.8%, giving it a comfortable cushion.

2.5 Liquidity Constraints: The 30-Day Survival Fund

Even a solvent bank can fail if it runs out of cash. Lehman Brothers in 2008 was arguably solvent on paper the day it filed for bankruptcy — it simply could not roll over its overnight borrowings fast enough. Post-crisis regulators introduced two liquidity ratios to prevent this.

LCR — Liquidity Coverage Ratio:

$$\mathrm{LCR}(t) = \frac{\mathrm{HQLA}(t)}{\mathrm{Net_Cash_Outflows}_{30d}(t)} \geq 1.0 \tag{10}$$

In plain terms

LCR asks: if a stress event triggered your worst-case deposit outflows and funding withdrawals for 30 days straight, would you have enough liquid assets (cash, Treasuries, central bank reserves) to survive without external help? You must hold enough High-Quality Liquid Assets (HQLA) to cover 100% of those outflows.

HQLA earns lower returns than loans. Cash in a reserve account earns the Fed Funds rate (5.25% today), while Treasuries earn 4.5–4.7%, versus corporate loans at 6.5%+.

Every dollar held in HQLA is a dollar that cannot be lent out at a higher yield. This is the "liquidity drag" — and minimizing it while staying compliant is one of the key optimization problems.

Net cash outflows in a 30-day stress scenario:

$$\text{Net_Cash_Outflows}_{30d}(t) = \sum_j \alpha_j \cdot L_j(t) - \text{Cash_Inflows}(t) \tag{11}$$

where α_j is the 30-day stressed runoff rate for each liability type: retail savings deposits have a low runoff (3%) because ordinary depositors are slow to flee; wholesale unsecured funding has a high runoff (25%) because professional investors move fast.

NSFR — Net Stable Funding Ratio:

$$\text{NSFR}(t) = \frac{\text{ASF}(t)}{\text{RSF}(t)} \geq 1.0 \tag{12}$$

In plain terms

NSFR is the 1-year version of LCR. It asks: are you using stable, long-term money (savings deposits, long-term bonds) to fund your long-term assets (mortgages, corporate loans)? Or are you dangerously using overnight borrowing to fund 30-year mortgages? NSFR forces the match.

Available Stable Funding (ASF) is what the bank raises with a stability weighting β^{ASF}; Required Stable Funding (RSF) is what its assets demand:

$$\text{ASF}(t) = \sum_j \beta_j^{\text{ASF}} \cdot L_j(t), \qquad \text{RSF}(t) = \sum_k \beta_k^{\text{RSF}} \cdot A_k(t) \tag{13}$$

Real World

Silicon Valley Bank collapsed in March 2023 partly because it funded long-dated bonds (high RSF) with short-term, rate-sensitive deposits from tech startups (low ASF in a stress scenario). When interest rates rose sharply, its bond portfolio lost value while deposit outflows accelerated. NSFR is designed specifically to force banks to avoid this mismatch.

2.6 The DFAST Stress Constraint: Surviving a Two-Year Recession

Every year, the Federal Reserve runs a "stress test" called DFAST (Dodd-Frank Act Stress Test) on large banks. The test asks: if the economy plunged into a severe recession — GDP falling 6%, unemployment reaching 10%, house prices dropping 25% — would this bank still be standing at the end of nine quarters?

The constraint this imposes on the optimization:

$$\min_{t \in [1,9]} \text{CET1_ratio}^{\text{sev_adv}}(t) \geq \kappa_{\text{floor}} \tag{14}$$

In plain terms

Even in the worst-case nine-quarter scenario fed to it by the Fed, the bank's capital ratio must not fall below the floor (Apex's internal floor is 6.5%; the regulatory minimum is 4.5%). This means the bank cannot grow its loan book so aggressively today that a recession two years from now would wipe out its capital cushion.

This constraint is what makes the problem non-trivial. The current composition of the balance sheet affects the *future* stressed capital path. A bank that lends heavily to commercial real estate today will have a much lower stressed CET1 trough if commercial real estate collapses — and the DFAST model will catch this and constrain the lending decision today.

2.7 Concentration Constraints: Do Not Put All Eggs in One Basket

If a bank lends half its capital to a single company, it is fine until that company defaults. Concentration rules prevent this:

$$A_{\text{counterparty-}i}(t) \leq 0.25 \cdot \text{CET1}(t), \qquad \forall\, i \tag{15}$$

No single counterparty can represent more than 25% of CET1. Sector concentration is also capped:

$$\sum_{i \in \text{sector-}s} A_i(t) \leq 0.25 \cdot \text{Total_Assets}(t), \qquad \forall\, s \tag{16}$$

2.8 The Complete Problem at a Glance

Assembling all constraints into one statement:

$$\begin{aligned}
\max_{\Delta A, \Delta L} \quad & \sum_{t=0}^{T} \delta^t \left[\text{NII}(t) - \text{CreditLoss}(t) - \text{OpRisk}(t)\right] \\
\text{s.t.} \quad & (2) : \text{Balance sheet identity (assets = liabilities + equity)} \\
& (7) : \text{CET1 ratio} \geq 7\% \\
& (9) : \text{Leverage ratio} \geq 3\% \\
& (10) : \text{30-day liquidity buffer} \geq 100\% \\
& (12) : \text{1-year stable funding} \geq 100\% \\
& (14) : \text{Stress-test CET1 trough} \geq 6.5\% \\
& (15) : \text{Single-name and sector concentration limits} \\
& A_k(t) \geq 0, \quad L_j(t) \geq 0 \quad \forall\, k, j, t
\end{aligned} \tag{17}$$

This is a multi-period nonlinear program (NLP) with integer components (lending decisions), stochastic elements (credit losses depend on macro realization), and regulatory feedback (RWA changes with asset mix). In practice, no bank solves this giant problem in one shot. Instead, it is decomposed into subproblems — capital optimization, liquidity

optimization, NII and duration management, and pricing — each handled by a specialist team with its own models. The next four sections walk through each subproblem.

3 Capital Optimization: Getting the Most From Every Dollar of Equity

Equity capital is the most expensive form of funding a bank has. Shareholders expect returns of 10–14% per year. The bank therefore wants to hold only as much equity as it absolutely needs, and deploy every dollar of it into the highest-returning activities.

3.1 The Capital Efficiency Frontier: A Map of Returns vs. Risk

Every business activity a bank undertakes — a mortgage, a corporate bond, an interest rate swap — earns a return and consumes capital. The *capital efficiency frontier* maps these activities by the return they generate per unit of capital consumed. The metric is RAROC:

$$\mathrm{RAROC}(k) = \frac{r_k^A - r_k^{\mathrm{FTP}} - \mathrm{EL}_k}{\mathrm{EC}_k} \tag{18}$$

where:

- r_k^A: gross yield on asset class k
- r_k^{FTP}: the internal "transfer price" charged to account for the cost of funding (Section 6)
- $\mathrm{EL}_k = \mathrm{PD}_k \cdot \mathrm{LGD}_k$: expected annual credit loss (probability of default × loss given default)
- EC_k: economic capital per unit of the activity

In plain terms

RAROC answers the question: after funding costs, after expected loan losses, how much does this activity earn per dollar of capital locked up? If RAROC is higher than the shareholder hurdle rate, it creates value. If not, the bank would do better returning the capital to shareholders.

A business line creates value if $\mathrm{RAROC}(k) > h$ (the hurdle rate, typically 11% for a money-center bank). Optimization re-allocates the balance sheet away from sub-hurdle activities toward high-RAROC ones.

3.2 Economic Capital: The Bank's Internal Risk Measurement

Beyond regulatory capital (which follows a rulebook), banks compute *economic capital* — their internal view of how much equity they truly need to absorb unexpected losses at a 99.9% confidence level.

$$\mathrm{EC}_{\mathrm{portfolio}} = \mathrm{VaR}_{99.9\%}(\mathrm{CreditLoss}_{1yr}) - \mathrm{EL} \tag{19}$$

In plain terms

At 99.9% confidence, the bank should only be wiped out once in 1,000 years. Economic capital is the extra cushion above expected losses that provides this protection. It is the bank's own honest assessment of risk — which may be more or less than the regulator's prescribed minimum.

The clever insight is that adding a loan to a *diversified* portfolio consumes less capital than adding the same loan in isolation, because it does not move in lockstep with other loans:

$$\mathrm{EC}_i = \frac{\partial \mathrm{EC}_{\text{portfolio}}}{\partial w_i} = \mathrm{Corr}(L_i, L_{\text{portfolio}}) \cdot \sigma_{L_i} \cdot z_{99.9\%} \tag{20}$$

Analogy

This is exactly the same logic as portfolio diversification in stock investing. Adding a tech stock to a tech-heavy portfolio barely reduces your total risk. Adding an oil stock to that same portfolio reduces it significantly — because when tech is down, oil often is not. A bank with a diversified loan book across sectors and geographies needs less capital than one concentrated in a single sector.

3.3 IMA vs. SA: The Capital Arbitrage Worth 30–60%

For trading book assets, banks can choose between two methods of calculating capital:

$$\mathrm{RWA}^{\mathrm{SA}}(k) = \rho_k^{\mathrm{SA}} \cdot \mathrm{EAD}_k \tag{21}$$

$$\mathrm{RWA}^{\mathrm{IMA}}(k) = 12.5 \cdot \left[\max\left(\mathrm{VaR}_t, \bar{k} \cdot \overline{\mathrm{VaR}}_{60d}\right) + \max\left(\mathrm{SVaR}_t, \bar{k} \cdot \overline{\mathrm{SVaR}}_{60d}\right)\right] \tag{22}$$

The SA (Standardised Approach) uses fixed, conservative risk weights prescribed by regulators. The IMA (Internal Models Approach) allows the bank to use its own VaR model, which — for a well-diversified book — typically computes 30–60% less capital than the SA.

The trade-off: IMA requires building, validating, and maintaining sophisticated risk models under SR 11-7 governance. SA is simpler but expensive. Every major bank with a significant trading book invests heavily in IMA — the capital saving pays for itself many times over.

3.4 Keeping Buffers: How Much Capital Is Too Much?

Apex runs a CET1 ratio of 13% against a regulatory minimum of 7%. Why hold six percentage points of "excess" capital? The answer is layered:

$$\mathrm{CET1}_{\text{target}} = \kappa_{\min} + B_{\mathrm{CCB}} + B_{\mathrm{DFAST}} + B_{\text{management}} \tag{23}$$

- $\kappa_{\min} = 4.5\%$: Basel III minimum — the absolute floor.

- $B_{\text{CCB}} = 2.5\%$: Capital Conservation Buffer. Breach this, and regulators restrict dividend payments and buybacks.
- $B_{\text{DFAST}} \approx 2.2\%$: The gap between today's CET1 and the projected trough under the Fed's severely adverse scenario. Consume this and the stress test submission fails.
- $B_{\text{management}} = 1.0$–$1.5\%$: Discretionary buffer. For strategic acquisitions, unexpected shocks, and the ability to move quickly without calling an emergency board meeting.

Real World

In the 2020 COVID shock, banks that had maintained substantial management buffers were able to absorb the initial wave of credit losses without cutting dividends or raising emergency equity. Banks that were running thin — close to their DFAST trough — had to freeze buybacks and in some cases accept government support. The management buffer is not idle excess capital. It is optionality.

4 Liquidity Optimization: Keeping the Lights On

A bank can be perfectly solvent — owning more than it owes — and still fail overnight if it cannot make its payments. This happened to Northern Rock in 2007, Bear Stearns in 2008, and nearly happened to dozens of others. Liquidity management is the discipline of ensuring the bank can always meet its obligations, even in a panic.

4.1 The Liquidity Drag: Paying for Peace of Mind

LCR and NSFR compliance requires holding a portfolio of HQLA. The "liquidity drag" is the yield it gives up relative to higher-yielding loans:

$$\text{Liquidity_Drag} = (\bar{r}^A - r^{\text{HQLA}}) \cdot \frac{\text{HQLA}}{A} \tag{24}$$

At Apex with \$45B of HQLA on a \$285B balance sheet:

$$\text{Liquidity_Drag} \approx (6.2\% - 4.5\%) \times 16\% = 27\,\text{basis points of NIM} \tag{25}$$

In plain terms

Apex gives up 0.27% of its entire earning-asset yield just to hold the required liquidity buffer. On a \$285B balance sheet, that is roughly \$770 million per year in forgone income. This is the price of never having a liquidity crisis.

The HQLA portfolio can itself be optimized — not all liquid assets are equally low-yielding. The optimization shifts from near-zero-yield excess cash at the Fed toward Treasuries and Agency bonds:

$$\max_{\text{HQLA mix}} \sum_{i \in \text{HQLA}} w_i \cdot r_i \quad \text{s.t.} \quad \begin{cases} \text{Level 1 assets} \geq 60\% \text{ of HQLA} \\ \text{Level 2A} \leq 40\% \text{ of HQLA} \\ \text{Level 2B} \leq 15\% \text{ of HQLA} \\ \text{Duration} \leq D_{\max} \\ \text{LCR} \geq 110\% \text{ (internal buffer)} \end{cases} \tag{26}$$

Replacing \$45B of excess cash with a laddered Treasury portfolio at 4.65% earns \$2.1B per year with *no reduction in LCR compliance*. That optimization alone is worth an analyst team.

4.2 Intraday Liquidity: The Hidden Plumbing

Separate from LCR/NSFR, the bank must fund each day's payment obligations as they arrive throughout the day. Apex processes billions of dollars of payments through Fedwire and CHIPS daily:

$$\text{Intraday_Peak}(t) = \max_{\tau \in [0, T_{\text{day}}]} \left[\text{Outflows}(\tau) - \text{Inflows}(\tau) - \text{Reserve}(\tau)\right]^+ \tag{27}$$

Tools for reducing the peak:

- Posting collateral to the Fed discount window (intraday credit at 0%)
- Payment sequencing — urgent payments go out at 9AM; non-urgent ones go at 3PM when inflows have arrived
- Correspondent banking netting — offsetting obligations with counterparties before settling

4.3 Building a Stable Funding Mix

The liability-side optimization aims for the lowest-cost, most stable funding mix that passes NSFR:

$$\min_{\Delta L_j} \sum_j r_j^L \cdot L_j \quad \text{s.t.} \quad \text{NSFR} \geq 100\%, \quad \text{LCR} \geq 100\% \tag{28}$$

The trade-off is fundamental:

- **Demand deposits** are nearly free (0.1% interest) and have high ASF if they are "core" (stable, relationship-driven) — but can evaporate quickly in a crisis.
- **Long-term unsecured bonds** are expensive (OIS + 80–120 bps) but have 100% ASF weight — maximum stability.

Definition 1 (Optimal Funding Tenor)**.** *Given the shape of the yield curve, the optimal funding maturity τ^* for a T-year asset satisfies:*

$$\tau^* = \arg\min_{\tau} \left[r^L(\tau) + \mathit{Rollover_Risk}(\tau, T)\right] \tag{29}$$

where $\mathit{Rollover_Risk}(\tau, T) = \sigma_r^2 \cdot (T - \tau)$ is the expected cost of refunding if short-term rates rise before maturity.

Real World

In 2021–2022, many regional banks made the mistake of funding long-dated mortgage portfolios with short-term, rate-sensitive deposits. When the Fed hiked rates by 525 bps in 18 months — the fastest hiking cycle in 40 years — those banks faced a brutal math problem: their loan income was locked in at 3% fixed rates while their funding costs rose toward 5%. The banks that survived were those that had locked in long-term funding when rates were low, accepting a small cost today to avoid a catastrophe later.

5 Asset-Liability Management: Playing Chess With Interest Rates

The single biggest driver of a bank's profitability is the level and direction of interest rates. When rates rise, banks that lend at floating rates make more money. When rates fall, banks with long-dated fixed-rate loans benefit. Asset-Liability Management (ALM) is the function that deliberately positions the bank to benefit from the rate environment — while managing the risk of being wrong.

5.1 The Repricing Gap: How Rate-Sensitive Is the Bank?

The repricing gap measures how much of the balance sheet will reprice at new interest rates within the next 12 months:

$$\Delta \text{NII} \approx \text{RSA}^{12m} \cdot \Delta r - \text{RSL}^{12m} \cdot \Delta r = \text{Gap}^{12m} \cdot \Delta r \tag{30}$$

where RSA^{12m} is the rate-sensitive asset balance (floating-rate loans, maturing fixed-rate loans) and RSL^{12m} is the corresponding liability balance.

In plain terms

If the bank has more floating-rate assets than floating-rate liabilities, a positive Gap, it benefits when rates rise (earns more on assets than it pays on liabilities). A negative Gap means the opposite — the bank benefits when rates fall. Most retail banks are asset-sensitive (positive Gap) because mortgages eventually reset while savings deposits are sluggish.

5.2 EVE: The Longer View of Rate Risk

NII sensitivity captures the next 12 months. But the bank's assets and liabilities also have market values that change with interest rates. Economic Value of Equity (EVE) captures this longer-horizon sensitivity:

$$\frac{\Delta \text{EVE}}{\text{EVE}} \approx -D_{\text{gap}} \cdot \frac{\Delta r}{1 + r} \tag{31}$$

where the duration gap is the difference between the average "life" of assets and the average "life" of liabilities, weighted by their sizes:

$$D_{\text{gap}} = D_A - \frac{L}{A} \cdot D_L \tag{32}$$

Analogy

Duration is financial physics for "how much does this asset's price change when rates move?" A 10-year bond's price falls much more when rates rise than a 1-year bond's price does. The duration gap measures whether the bank's assets change in value *more or less* than its liabilities when rates shift. A large positive duration gap means big EVE losses when rates rise — which is exactly what happened to SVB.

Proposition 1 (The Fundamental ALM Trade-off)**.** *A bank cannot simultaneously protect both NII and EVE against interest rate risk. Shortening asset duration (holding more floating-rate loans) protects NII in a rising rate environment but leaves EVE more exposed. Lengthening asset duration (more fixed-rate loans and long bonds) stabilizes EVE but makes NII vulnerable to rate rises.*

The optimal duration gap maximizes the weighted utility of NII and EVE outcomes across rate scenarios:

$$\max_{D_{\text{gap}}} \quad U(\Delta\text{NII}, \Delta\text{EVE}) = w_1 \cdot \mathbb{E}[\Delta\text{NII}] - w_2 \cdot \text{Var}(\Delta\text{NII}) - w_3 \cdot |\Delta\text{EVE}|^2 \tag{33}$$

subject to the risk appetite band $[D^-, D^+]$ set by the ALM Committee.

5.3 Hedging: Using Swaps to Adjust the Duration Gap

If the current duration gap is outside the target range, the ALM desk uses interest rate swaps to correct it without buying or selling physical assets:

- **Pay-fixed receiver IRS**: The bank pays a fixed rate and receives a floating rate — effectively converting a floating-rate liability into a fixed one, increasing liability duration.
- **Receive-fixed payer IRS**: The bank receives a fixed rate and pays a floating one — converting a fixed asset into a floating asset, reducing asset duration.

The notional of swap needed to close a duration gap of ΔD:

$$N_{\text{swap}} = \frac{\Delta D \cdot A}{D_{\text{swap}}} \tag{34}$$

In plain terms

Swaps let the bank change its interest rate exposure without actually restructuring its loan book — which would take months and cost real money. It is like adjusting the bet without changing the underlying assets.

5.4 Convexity: When Duration Alone Is Not Enough

Fixed-rate mortgages behave in an awkward way that makes them particularly tricky to hedge. When rates fall, homeowners refinance their mortgages early — returning the

bank's capital right when reinvestment rates are lowest. When rates rise, those same homeowners hold onto their cheap fixed-rate mortgages for as long as possible. This is *negative convexity*:

$$P(r) \approx P(r_0)\left[1 - D_{\text{eff}} \cdot \Delta r + \frac{1}{2} C_{\text{eff}} \cdot (\Delta r)^2\right] \tag{35}$$

For mortgage-backed securities (MBS), $C_{\text{eff}} < 0$ — meaning the price falls faster when rates rise *and* rises slower when rates fall than a simple bond would. Banks with large MBS portfolios hedge this with swaptions and callable bonds that provide offsetting positive convexity.

6 Fund Transfer Pricing: The Internal Economy of a Bank

Imagine a bank with 20 business lines. The mortgage desk lends \$40B at 4.5%. The corporate banking desk lends \$30B at 6.5%. Which is more profitable? The answer depends entirely on what it costs each desk to fund those loans. Without a rigorous internal pricing system, a bank cannot know where it is actually making money — and will misprice every product it sells.

Fund Transfer Pricing (FTP) is the solution: a system of internal charges and credits that transfers the cost (and benefit) of funding between business lines and a central Treasury pool.

6.1 How FTP Works

Every asset the bank creates generates an FTP *charge* to the originating desk. Every deposit raised generates an FTP *credit*:

$$\text{FTP_Charge}(k) = \underbrace{r^{\text{OIS}}(\tau_k)}_{\substack{\text{Risk-free}\\ \text{funding cost}}} + \underbrace{\lambda_k}_{\substack{\text{Liquidity}\\ \text{premium}}} + \underbrace{\psi_k}_{\substack{\text{Optionality}\\ \text{premium}}} \tag{36}$$

Analogy

Think of Treasury as the bank's internal bank. The mortgage desk wants to fund \$1B of 30-year mortgages? Treasury says: "I'll fund that for you at OIS plus our liquidity premium for 30-year money." The desk then knows the true all-in cost of its lending. If it cannot make a spread above that FTP charge, it is not earning its keep.

The three components of the FTP charge:

- $r^{\text{OIS}}(\tau_k)$: the risk-free rate at the matched maturity, from the OIS (overnight indexed swap) curve. This is the rock-bottom cost of risk-free money for this tenor.
- λ_k: the liquidity premium — the extra cost above OIS to raise actual term funding in the market. A 5-year bond today might price at OIS + 45bps.
- ψ_k: the optionality premium for embedded options. A mortgage with a prepayment option costs more to fund than a plain loan.

6.2 How FTP Eliminates the Illusion of Cheap Funding

Without FTP, banks consistently fool themselves. Here is the illusion:

A bank funds itself with overnight deposits at 0.5% and lends on 30-year fixed mortgages at 4.5%. Gross spread: 4%. Looks great. But the bank is not earning 4% — it is earning a tiny spread and accumulating enormous unhedged interest rate risk. If rates rise 2%, the deposit costs jump to 2.5% overnight while the mortgage is locked in at 4.5%. Real spread: 2%. And the bank now faces enormous duration gap losses.

FTP forces honest accounting:

$$\mathrm{NIM}_{\mathrm{adjusted}}(k) = r_k^A - \mathrm{FTP}(k) - \mathrm{EL}_k \tag{37}$$

A 30-year mortgage funded at the 30-year FTP rate (OIS + 45bps at that tenor) suddenly shows its true profitability — which may be far less exciting than the raw gross spread suggests.

6.3 The Components of the Liquidity Premium

$$\lambda = \lambda_{\mathrm{term}} + \lambda_{\mathrm{LCR}} + \lambda_{\mathrm{NSFR}} + \lambda_{\mathrm{CFP}} \tag{38}$$

- λ_{term}: observed from the bank's own bond issuance spread — how much above OIS does it actually pay to borrow for 5 years?
- λ_{LCR}: the cost of holding the HQLA buffer attributable to this liability type. A deposit with a high runoff factor forces the bank to hold more HQLA against it.
- λ_{NSFR}: the cost of providing stable funding to match the stable funding requirement of the asset.
- λ_{CFP}: the cost of maintaining Contingency Funding Plan reserves for stress scenarios.

6.4 Incentive Design: Why Getting FTP Wrong Is Costly

FTP rates shape behavior across the bank. Setting them wrong is dangerous:

- **Too low**: Loan desks over-originate because funding is underpriced. The balance sheet grows unsustainably. Treasury absorbs the real funding costs silently. This is exactly what happened at many European banks pre-2008.
- **Too high**: Business lines under-price deposit gathering. Treasury retains excess profits; front-office desks are cross-subsidizing Treasury's profit targets. Loan volumes shrink and clients take their business elsewhere.

FTP rates must be reviewed quarterly by the ALM Committee and recalibrated to market rates whenever the yield curve shifts materially.

7 RAROC: Should We Even Do This Trade?

Every time a banker wants to make a loan or execute a trade, someone in the risk function should be asking: given everything this activity costs in capital, funding, and expected losses, are we actually earning more than our cost of equity? This is the RAROC question.

7.1 The Full RAROC Formula

$$\mathrm{RAROC}(k) = \frac{(r_k^A - r_k^{\mathrm{FTP}}) \cdot A_k - \mathrm{EL}_k \cdot A_k - \mathrm{OpCost}_k}{\mathrm{EC}_k} \tag{39}$$

In plain terms

The numerator is: what the activity actually earns (after funding costs and expected losses and operating expenses). The denominator is: how much capital is locked up to support it. Divide the first by the second, and you have the true risk-adjusted return on the capital deployed.

Definition 2 (RAROC Hurdle Rate)**.** *The hurdle rate h is the bank's cost of equity — what shareholders expect as a minimum annual return. Using the Capital Asset Pricing Model:*

$$h = r_f + \beta \cdot (r_m - r_f) = 4.5\% + 1.2 \times 5.5\% = 11.1\% \tag{40}$$

for a money-center bank with market beta $\beta \approx 1.2$. Any business line with RAROC $<$ 11.1% is destroying shareholder value and should be restructured or exited.

7.2 The Minimum Loan Rate: A Step-by-Step Calculation

How low can a loan rate go before the bank loses money on a risk-adjusted basis? Here is the calculation for a 5-year commercial loan:

$$r^* = r^{\mathrm{FTP}}(\tau) + \lambda^{\mathrm{FTP}} + \mathrm{EL} + \frac{h \cdot \mathrm{EC}}{A} + \mathrm{OpCost} \tag{41}$$

Working through the numbers:

$$r^{\mathrm{OIS}}(5Y) = 4.40\% \quad \text{(5-year OIS swap rate)} \tag{42}$$

$$\lambda^{\mathrm{FTP}}(5Y) = 0.45\% \quad \text{(liquidity premium for 5-year funding)} \tag{43}$$

$$\mathrm{EL} = \mathrm{PD} \cdot \mathrm{LGD} = 1.5\% \times 45\% = 0.675\% \quad \text{(expected annual credit loss)} \tag{44}$$

$$\frac{h \cdot \mathrm{EC}}{A} = 11.1\% \times 8\% = 0.888\% \quad \text{(capital charge)} \tag{45}$$

$$\mathrm{OpCost} = 0.20\% \quad \text{(operations, compliance, origination)} \tag{46}$$

$$r^* \approx \mathbf{6.61}\% \tag{47}$$

In plain terms

Every component of this calculation represents a real cost. Lending below 6.61% on this loan means the bank is giving away shareholder capital without adequate compensation. Above 6.61%, the loan creates value. A banker who accepts a 6.0% rate because "the client relationship is important" is essentially asking shareholders to subsidize that relationship.

7.3 Capital Allocation: Distributing the Total Budget

The total economic capital must add up correctly:

$$\sum_k \mathrm{EC}_k = \mathrm{EC}_{\text{portfolio}} \tag{48}$$

Using Euler's theorem on homogeneous risk measures:

$$\mathrm{EC}_k = w_k \cdot \frac{\partial \mathrm{VaR}_\alpha}{\partial w_k} \tag{49}$$

This ensures that the diversification benefit is properly distributed across business lines — a concentrated desk that adds no diversification bears more of the total capital charge than a desk whose exposures offset other risks in the portfolio.

8 Stress Testing: Planning for the Unthinkable

Regulators require banks to prove they could survive a severe recession. But the best-run banks use stress tests not just for compliance — they use them as strategic planning tools. "We can grow loans by $30B," a banker might say. The DFAST model responds: "Yes, but only if you are comfortable watching your CET1 ratio fall to 7.1% in Q6 of the severely adverse scenario. Are you?"

8.1 DFAST as a Constraint on Business Strategy

The severely adverse CET1 trough has direct operational consequences:

- **Maximum loan growth capacity**: Adding $10B of new commercial real estate loans today increases stressed losses by approximately $\mathrm{EL}_{\text{stress}} \times \$10B/\mathrm{RWA}$ and reduces the trough CET1 accordingly. If the trough is already close to the floor, this growth is blocked.
- **Dividend capacity**: Every dollar of dividend paid reduces starting CET1 and directly lowers the trough. Banks announce dividend plans only *after* running DFAST.
- **Share buyback capacity**: Buybacks above the buffer could cause the trough to breach the internal floor. The CFO and CRO jointly approve buyback programs with DFAST sensitivity analysis.

8.2 Reverse Stress Testing: Finding the Failure Modes

Forward DFAST asks: "what happens in the Fed's scenario?" Reverse stress testing asks: "what would have to happen to actually break this bank?"

$$\min_{\text{scenario}} \text{CET1_trough(scenario)} \quad \text{s.t.} \quad \text{Scenario plausible under risk model} \tag{50}$$

The three most dangerous scenarios for Apex:

1. **Credit concentration collapse**: A 55% drop in commercial real estate values combined with a deposit run creates simultaneous asset losses and liability instability. Each reinforces the other in a downward spiral.

2. **The Duration Trap** (SVB redux): A rapid +300bps rate shock reduces EVE close to zero while simultaneously driving up funding costs on short-dated liabilities. SVB became the poster child for this scenario in March 2023.
3. **The Liquidity-Solvency Spiral**: An LCR breach forces fire-sales of HQLA assets — but those forced sales depress HQLA asset prices, which makes the LCR breach worse, which forces more sales. A liquidity problem becomes a solvency problem.

Each scenario generates a "distance to failure" metric:

$$D_{\text{failure}} = \frac{\text{CET1_current} - \text{CET1_at_failure}}{\text{CET1_1yr_volatility}} \tag{51}$$

The higher this distance, the more resilient the bank is. Maintaining large distances on all three failure modes simultaneously is the goal of strategic balance sheet optimization.

9 Technology: The Calculators That Power It All

A balance sheet optimization program is only as good as its models. Building and maintaining the calculator stack is a multi-year engineering program requiring quants, technologists, and governance specialists working in concert.

9.1 The Minimum Production Calculator Stack

Table 2: Core Calculator Stack for Balance Sheet Optimization

Calculator	What It Does	Key Output
RWA / Capital Engine	Computes regulatory capital for all assets	CET1/Tier1/Total ratios
VaR / SVaR Engine	Market risk capital (IMA method)	10-day 99% VaR
FRTB-SA Engine	Standardised approach as floor	FRTB-SA capital
IFRS 9 ECL Engine	Expected credit losses, loan staging	Provision and ECL ratio
DFAST Engine	9-quarter recession simulation	CET1 trough
ALM Engine	Rate sensitivity, duration, NII	Duration gap, NIM
FTP Engine	Internal funding charges	Desk P&L (adjusted)
Collateral / SIMM	Margin requirements under ISDA	IM requirement
XVA Suite	Full cost of derivatives trades	CVA/DVA/FVA/KVA
RAROC Calculator	Minimum viable pricing	Loan floor rate
LCR / NSFR Calculator	Liquidity ratio compliance	LCR, NSFR ratios
Concentration Monitor	Exposure limits by name/sector	Breach flags

9.2 The Data Architecture

All twelve calculators consume data from the same underlying infrastructure:

- **Positions layer**: Real-time feed from OMS (Order Management System) and loan systems. All instruments mapped to ISIN/CUSIP. At Apex: 500ms tick interval for trading book positions.
- **Market data layer**: Live prices, interest rate curves, implied volatility surfaces, credit spreads. The VaR engine re-runs every time a new market data tick arrives.

- **Reference data**: CSA terms (for collateral), counterparty credit ratings, ISDA SIMM risk weights, FATF grey-list for AML.
- **Event log**: Append-only audit trail of all balance sheet actions, capital calls, and model outputs. Regulators expect this. Apex uses SQLite with a write-ahead log.
- **Regulatory parameters**: Fed-published DFAST scenarios, Basel-prescribed risk weights, LCR runoff factors. Updated annually.

9.3 How Much Computing Power Does This Require?

Table 3: Computational Requirements by Calculator

Calculator	Frequency	Run Time	CPU Cores	Memory
VaR MC (2,000 paths, 11 instruments)	Daily	50ms	1	50MB
FRTB-SA (delta/vega/curvature)	Daily	200ms	4	200MB
DFAST (9 quarters, 3 scenarios)	Quarterly	5–30s	8	1GB
XVA (1,000 paths, 50 trades)	On each new trade	2–10s	16	4GB
SIMM (all CSAs)	Daily	100ms	2	100MB
LCR / NSFR	Daily	500ms	2	200MB
ALM / NII sensitivity	Monthly	1s	2	100MB
IFRS 9 ECL (50 obligors)	Quarterly	5ms	1	20MB

For a production-scale institution (850+ obligors, 10,000+ derivative trades, full FRTB-IMA), the requirements scale dramatically:

- **XVA**: 64–128 GPU cores for real-time Monte Carlo (using AAD — Adjoint Algorithmic Differentiation — for sensitivities).
- **FRTB IMA**: Historical simulation on 10+ years of daily data across 500+ risk factors. Petabytes of historical market data.
- **DFAST**: Cluster parallelization — 3 scenarios $\times$ 9 quarters $\times$ multiple credit models $\times$ macro satellite models.

9.4 The Dashboard Suite: Making It Visible

The models are only useful if decision-makers can see their outputs clearly and act on them. A best-practice balance sheet optimization dashboard suite:

1. Capital Dashboard. CET1, Tier 1, Total Capital, and Leverage ratios plotted against regulatory floors and internal targets. RWA decomposed by risk type. Capital headroom to each constraint. An 8-quarter trailing history so trends are visible. The CRO sees this before the markets open every morning.

2. Liquidity Dashboard. LCR and NSFR ratios with intraday updates for LCR. HQLA composition waterfall (Level 1 vs. 2A vs. 2B). 30-day net cash outflow breakdown. Intraday liquidity utilization against credit lines. A red alert if LCR drops below 110% (internal buffer) at any point during the day.

3. ALM / Interest Rate Risk Dashboard. NII sensitivity bar charts under ±100bps and ±200bps shocks. EVE sensitivity. Repricing gap schedule across 7 time buckets. Duration gap trend versus the risk appetite band. An SVB-style warning flag if EVE sensitivity exceeds –15% under an instantaneous +200bps shock.

4. DFAST / Stress Dashboard. Nine-quarter CET1 trajectory under all three Fed scenarios. The chart shows the minimum floor line and the internal risk appetite band as horizontal reference lines. Loss attribution waterfall (how much came from credit losses vs. market losses vs. operational risk vs. PPNR weakness). Distance to the regulatory minimum.

5. FTP / RAROC Dashboard. Business-line ranking by FTP-adjusted P&L. RAROC versus hurdle rate for each desk. A heat-map of new origination pricing versus the RAROC floor. The FTP curve visualized as OIS plus liquidity premium at each tenor. Any desk below the hurdle rate shows in red.

6. XVA Dashboard. CVA, DVA, FVA, MVA, ColVA, and KVA broken down by counterparty and trade. Live WebSocket updates as new trades are submitted to the OMS. XVA P&L explain versus prior day (how much of today's XVA change came from market moves, credit spread changes, or new trades).

7. Boardroom / Scenario Dashboard. Real-time narrative from the AI agent layer. Scenario engine with live risk factor injection. Model registry showing the governance status of every quantitative model. The forum where the CRO, CFO, and Head of Trading discuss balance sheet strategy.

10 How Balance Sheet Optimization Actually Makes Money

Every section so far has described constraints and costs. Now comes the payoff: what does all of this discipline actually generate in revenue?

10.1 Lever 1 — Liberating Trapped Capital

Capital is trapped when it is supporting assets that consume disproportionate regulatory risk weight relative to the income they generate. Releasing it is pure value creation.

Example. Apex holds \$10B of lower-grade commercial real estate loans (risk weight 100%, yield 6%). These can be swapped — through secondary market sales and new origination discipline — for investment-grade corporate bonds (risk weight 20%, yield 5.5%):

$$\text{RWA reduction} = (100\% - 20\%) \times \$10B = \$8B \tag{52}$$

$$\text{CET1 freed} = 7\% \times \$8B = \$560M \tag{53}$$

$$\text{NII sacrifice} = (6\% - 5.5\%) \times \$10B = \$50M/\text{yr} \tag{54}$$

The trade gives up \$50M/yr in income but frees \$560M of equity capital. If that capital can be redeployed — say, into a derivatives business generating RAROC 12% on \$560M — it generates \$67M, net benefit \$17M/yr from a simple asset swap.

10.2 Lever 2 — Positioning the Duration Gap

The most powerful lever in ALM is simply having the right duration gap going into a rate cycle.

Example. Apex enters 2022 with a duration gap of +0.41 years and starting NII of \$40B/year. The Fed hikes 525bps over 18 months. Result:

$$\Delta\text{NII} = \text{Gap}^{12m} \times \Delta r \approx +\$6.8B/\text{yr} \tag{55}$$

In plain terms

A bank with a zero duration gap would capture nothing from the hiking cycle. A bank over-hedged to a negative gap would actually *lose* NII as rates rose. Apex's +0.41yr gap, maintained through disciplined ALM, generated an extra \$6.8B per year in NII while most peers scrambled to manage their inverted net interest margins.

The other side of the bet: if rates had instead fallen 200bps, Apex would have given up \$5.9B in NII. The gap positioning is not a free lunch — it is a deliberate rate view. The discipline is in choosing the size of the bet and not over-concentrating.

10.3 Lever 3 — Turning HQLA Into a Revenue Centre

Holding \$45B in emergency liquidity is mandatory. But *how* you hold it is a choice. The difference between holding it all in zero-yield Fed reserves versus a duration-managed Treasury and Agency portfolio:

$$\text{HQLA yield gain} = (4.65\% - 0\%) \times \$45B = \$2.1B/\text{yr} \tag{56}$$

In plain terms

This is not a theoretical gain. Banks that parked hundreds of billions in excess reserves at the Fed from 2010 to 2021 while rates were zero sacrificed almost nothing. But as reserves accumulated in the post-COVID quantitative easing era, those same banks could have held Treasuries at 4.5% instead — and earned over \$2 billion per year with no additional risk.

10.4 Lever 4 — XVA Pricing: Charging the True Cost of Derivatives

When a bank enters into a derivatives trade with a client, the trade costs the bank capital (for regulatory requirements), funding (for posting initial margin), and carries credit risk (if the counterparty defaults). These costs are bundled into XVA adjustments.

The KVA (Capital Valuation Adjustment) is the present value of the capital cost over the trade's lifetime:

$$\text{KVA} = h \cdot \int_0^T \mathbb{E}[\text{EC}(t)] \cdot \text{df}(t)\, dt \tag{57}$$

In plain terms

KVA answers: "If I have to hold capital against this trade every day for the next 5 years, what is the total present value of what that capital *could* have earned if deployed elsewhere?" Pricing KVA into the trade upfront means the desk is compensated for that long-dated capital consumption the moment the trade is done. Banks that do not price KVA look profitable today but are accumulating invisible long-dated liabilities.

Typical KVA is 5–15 basis points on notional for a 5-year IRS. On a \$500B derivatives book, that is \$250–750M per year in value correctly priced into trades.

10.5 Lever 5 — Clearing and Regulatory Arbitrage

Interest rate swaps and credit default swaps can be executed either bilaterally (between the bank and a single counterparty) or via a central clearing house (LCH, CME). Cleared derivatives attract dramatically lower initial margin requirements:

$$\text{IM (bilateral SIMM)} \approx \$70M \tag{58}$$

$$\text{IM (LCH cleared)} \approx \$25M \tag{59}$$

$$\text{IM saving} = \$45M \tag{60}$$

At a funding cost of 5.25%, the \$45M margin saving generates $\$45M \times 5.25\% = \$2.4M$ per year in FVA/MVA savings on *this single portfolio*. Scaled to a \$500B notional derivatives book, the clearing-driven IM reduction saves \$50–150M per year.

10.6 The Full Revenue Picture

Table 4: Estimated Annual Revenue Impact (Apex Scale, \$285B Balance Sheet)

Optimization Lever	Annual P&L Impact	Key Mo
HQLA yield optimization	+\$2.1B	ALM Eng
NII duration positioning (+200bps scenario)	+\$6.8B	ALM Eng
Clearing-driven IM reduction (MVA/FVA saving)	+\$50–150M	SIMM / X
KVA pricing into new derivatives	+\$100–250M	XVA S
CET1 release via lower-RWA assets	+\$200–500M (redeployed)	Capital Eng
Loan pricing discipline (RAROC floor)	Prevents erosion of 50–100bps NIM	RAR
FTP governance (correct term premium)	Eliminates \$0.5–1.5B cross-subsidy	FTP Eng

The cumulative uplift from a mature balance sheet optimization program — relative to an unoptimized baseline where a bank simply grows loans and ignores these levers — is typically 15–40 basis points on total assets, or \$0.4–1.1B per year at Apex's scale.

At JPMorgan's $3.9 trillion balance sheet, a mature optimization program generates an estimated $3–8 billion per year in incremental returns.

To put that in perspective: the entire engineering, quant research, and technology investment needed to build and maintain the calculator stack described in Section 9 costs perhaps $200–400M per year all-in. The return on that investment, just from the HQLA optimization and FTP governance alone, is 5–10x.

11 Governance: Keeping the Machine Honest

The most sophisticated optimization program is worthless if the models driving it are wrong and nobody notices. SR 11-7 model risk governance exists precisely to prevent this.

11.1 Every Model Needs an Independent Reviewer

Under the Federal Reserve's SR 11-7 guidance, every quantitative model used in a material risk or capital decision must be:

- **Documented** in a Model Development Document (MDD): what assumptions it makes, what it cannot do, who is allowed to use it and for what.
- **Independently validated** by a Model Validation Officer who did not build the model — replicating key calculations, checking the math, and stress-testing the assumptions.
- **Monitored**: model KRIs (Key Risk Indicators) tracked monthly. VaR exception rates. SIMM vs. actual IM reconciliation. DFAST vs. actual CET1 variance. ALM model vs. actual NII error.
- **Use-authorized**: every model's Section 7 specifies what it can be used for, what it cannot be used for, and who may use it.

11.2 The ALM Committee: Where Strategy Gets Made

The ALM Committee is the governing body for all balance sheet optimization decisions. Membership:

- CRO (Chair)
- CFO / Treasurer
- Head of Trading
- Head of Commercial Banking
- Head of Model Validation (advisory)
- Chief Risk Economist (macro forecasting)

The Committee meets monthly and reviews: NII sensitivity, EVE sensitivity, duration gap, LCR/NSFR, FTP rates, and any proposed balance sheet transaction above materiality thresholds. No major balance sheet move happens without the Committee's sign-off.

11.3 The Three Lines of Defence

No single team can both operate a model and audit it objectively. The three-lines framework separates these roles clearly:

1. **First line (Treasury / business lines)**: Build and run the models. Own the P&L implications. Bring proposals to the ALM Committee.
2. **Second line (Risk / Model Validation)**: Independent of the first line. Cannot be overruled by them. Validates models, sets risk limits, monitors KRIs, and challenges assumptions.
3. **Third line (Internal Audit)**: Annual independent review of whether the governance framework itself is working. Tests that the first and second lines are actually doing what they claim.

12 Conclusion: The Highest-Leverage Discipline in Banking

You began this document without a financial background. Let us close with the core intuitions you should take away.

A bank is simultaneously solving several hard problems at once: earning a return on capital, surviving a 30-day liquidity crisis, surviving a two-year recession, not betting the house on a single borrower, and doing all of this while navigating a dense maze of regulation. Balance sheet optimization is the discipline of navigating all of those constraints simultaneously — and finding, within the feasible space, the position that earns the most.

The key insight that separates world-class practitioners from average ones is this: **the regulatory constraints are not obstacles to profitability — they are the map of the efficient frontier.** A bank that treats capital ratios, LCR, and DFAST as boxes to tick will always earn less than one that treats them as inputs to a rigorous optimization.

A 7% CET1 floor is not just a compliance number. It tells you precisely how much risk-weighted lending capacity you have. An NSFR requirement is not paperwork. It forces you to match asset maturities with liability maturities, which eliminates the most dangerous source of bank failure. A DFAST trough projection is not just a regulatory filing. It is the forward-looking constraint that prevents a bank from growing into a future recession it cannot survive.

Banks that truly internalize these constraints — building them into every loan pricing decision, every derivative trade, every funding transaction — consistently outperform peers on both profitability and resilience. At Apex Global Bank, the infrastructure described in this document provides the quantitative foundation for that program.

The next frontier is continuous, real-time balance sheet optimization: moving from a quarterly planning cycle to a model-driven optimization that updates recommendations as market conditions change. This requires AAD (Adjoint Algorithmic Differentiation) sensitivities across all models, GPU-accelerated Monte Carlo, and reinforcement learning for the hedging decision layer — capabilities commercially available and being deployed at G-SIBs today. The banks that build this capability next will have a structural advantage in the decade ahead.

12 Glossary of Abbreviations

Abbreviation	Full Term and Brief Description
AAD	**Adjoint Algorithmic Differentiation.** A mathematical technique for computing derivatives of model outputs with respect to all inputs simultaneously. Used in XVA and FRTB for computing "Greeks" (sensitivities) at GPU speed.
AFS	**Available for Sale.** A balance sheet classification for securities the bank intends to hold but may sell before maturity. Unrealised gains and losses flow through Other Comprehensive Income (OCI).
ALM	**Asset-Liability Management.** The discipline of managing the interest rate sensitivity and liquidity profile of the entire balance sheet. Overseen by the ALM Committee.
AML	**Anti-Money Laundering.** Regulatory framework and internal controls designed to detect and report suspicious financial activity. Governed by the Bank Secrecy Act and FinCEN guidance in the US.
ASF	**Available Stable Funding.** The numerator of the NSFR. Liabilities weighted by their stability over a one-year stress horizon — long-term debt receives a high ASF factor; overnight funding receives a low one.
BCBS	**Basel Committee on Banking Supervision.** The international standard-setter for bank prudential regulation, hosted at the Bank for International Settlements (BIS) in Basel, Switzerland.
CAPM	**Capital Asset Pricing Model.** A finance model that relates expected return to systematic risk (beta). Used to estimate the bank's cost of equity for the RAROC hurdle rate calculation.
CCB	**Capital Conservation Buffer.** A 2.5% CET1 buffer above the 4.5% minimum. Breaching it triggers restrictions on dividends and buybacks. Together they set the effective CET1 floor at 7%.
CCP	**Central Clearing Counterparty.** An exchange-like intermediary (e.g. LCH, CME) that sits between the two sides of a derivatives trade, guaranteeing settlement and netting obligations. Attracts lower initial margin than bilateral trades.
CET1	**Common Equity Tier 1.** The highest-quality form of bank capital: ordinary shares plus retained earnings minus regulatory deductions. The headline capital ratio reported by banks and scrutinised by regulators.
CFO	**Chief Financial Officer.** The senior executive responsible for the bank's financial planning, reporting, and capital management.

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Abbreviation	Full Term and Brief Description
CFP	**Contingency Funding Plan.** A documented strategy for how the bank would raise emergency liquidity in a stress scenario. The cost of maintaining CFP reserves is a component of the FTP liquidity premium.
CHIPS	**Clearing House Interbank Payments System.** The US private large-value payment system handling the majority of USD cross-border and domestic wholesale transactions.
CME	**Chicago Mercantile Exchange.** One of the world's largest derivatives exchanges and clearing houses.
CMS	**Constant Maturity Swap.** An interest rate swap where one leg pays a floating rate linked to a constant maturity rate (e.g., the 10-year swap rate) rather than a standard overnight rate.
CRE	**Commercial Real Estate.** Loans and securities backed by income-producing properties (offices, retail, warehouses). High risk weight; a common source of credit concentration risk.
CRIF	**Common Risk Interchange Format.** The standardised file format used to exchange SIMM sensitivity data between counterparties under the ISDA SIMM methodology.
CRO	**Chief Risk Officer.** The senior executive responsible for the bank's risk management framework, including capital, credit, market, and operational risk.
CSA	**Credit Support Annex.** A legal agreement appended to an ISDA Master Agreement that governs the posting and receipt of collateral (both variation margin and initial margin) between two derivatives counterparties.
CUSIP	**Committee on Uniform Securities Identification Procedures.** A nine-character alphanumeric identifier for North American financial instruments.
CVA	**Credit Valuation Adjustment.** The market value of counterparty credit risk embedded in a derivatives portfolio. Represents the expected loss from counterparty default, discounted to present value.
DFAST	**Dodd-Frank Act Stress Test.** An annual Federal Reserve supervisory exercise requiring US banks to model their capital adequacy under three macroeconomic scenarios (baseline, adverse, and severely adverse) over nine quarters.
DRC	**Default Risk Charge.** A component of FRTB-SA capital that captures the jump-to-default risk for credit instruments not captured by the sensitivity-based method.
DV01	**Dollar Value of 01 (one basis point).** The change in market value of a fixed-income instrument for a 1 basis point (0.01%) parallel shift in interest rates. A fundamental measure of interest rate risk.

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Abbreviation	Full Term and Brief Description
DVA	**Debit Valuation Adjustment.** The adjustment to derivatives value reflecting the bank's *own* credit risk — the value the bank would capture if it defaulted on its derivatives obligations.
EAD	**Exposure at Default.** The estimated outstanding balance of a loan or derivative at the moment a counterparty defaults. Used in credit risk capital calculations: RWA = risk weight × EAD.
EC	**Economic Capital.** The bank's internal estimate of the capital needed to absorb unexpected losses at a given confidence level (typically 99.9%). Distinct from regulatory capital, which follows a fixed rulebook.
ECL	**Expected Credit Loss.** Under IFRS 9, the probability-weighted estimate of credit losses over the life of a financial instrument. Drives loan loss provisioning.
EL	**Expected Loss.** EL = PD × LGD × EAD. The average annual credit loss on a loan or portfolio, incorporated into pricing via the RAROC formula.
EVE	**Economic Value of Equity.** The net present value of all asset and liability cash flows. EVE sensitivity measures how much equity value is gained or lost under a shift in interest rates.
FATF	**Financial Action Task Force.** The inter-governmental body that sets international standards for combating money laundering and terrorist financing. Maintains the "grey list" of jurisdictions under increased monitoring.
Fed	**Federal Reserve.** The central bank of the United States; the primary prudential regulator for large US bank holding companies.
FHLB	**Federal Home Loan Bank.** A US government-sponsored enterprise that provides wholesale funding (advances) to member banks, primarily used for mortgage funding.
FinCEN	**Financial Crimes Enforcement Network.** A bureau of the US Treasury that administers the Bank Secrecy Act and manages AML/CFT reporting.
FRTB	**Fundamental Review of the Trading Book.** The Basel IV reform of market risk capital standards (finalized 2019, effective 2025). Introduces a revised SA floor and tighter IMA eligibility rules.
FTP	**Fund Transfer Pricing.** An internal pricing mechanism that allocates the cost and benefit of funding to each business line, ensuring desks see the true all-in cost of originating assets and raising liabilities.

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Abbreviation	Full Term and Brief Description
FVA	**Funding Valuation Adjustment.** The cost of funding uncollateralised derivatives exposures. Reflects the bank's own credit spread applied to the unsecured funding component of the derivatives book.
GDP	**Gross Domestic Product.** The total monetary value of goods and services produced in an economy. A primary driver variable in DFAST macroeconomic scenario design.
G-SIB	**Global Systemically Important Bank.** Banks designated by the FSB as posing systemic risk to the global financial system. Subject to additional capital surcharges (1–3.5% of RWA) and enhanced supervision.
GPU	**Graphics Processing Unit.** Massively parallel processor originally designed for graphics rendering. Widely adopted for Monte Carlo simulation in XVA and VaR, where thousands of paths can run simultaneously.
HQLA	**High-Quality Liquid Assets.** Assets that can be quickly converted to cash with minimal loss of value in a stress scenario. Level 1 (cash, Treasuries), Level 2A (Agency debt, covered bonds), and Level 2B (equities, corporate bonds) receive progressively lower credit in LCR.
HTM	**Held to Maturity.** A balance sheet classification for securities the bank intends to hold until they mature. Not marked to market — which is why SVB's HTM bond losses did not immediately appear in its capital ratios.
IFRS 9	**International Financial Reporting Standard 9.** The accounting standard governing the classification and measurement of financial instruments, including the three-stage expected credit loss model for loan provisioning.
IM	**Initial Margin.** Collateral posted at the outset of a derivatives trade (and dynamically adjusted) to cover potential future exposure. Computed under SIMM for bilateral trades; exchange-defined for cleared trades.
IMA	**Internal Models Approach.** The Basel/FRTB method that allows banks with regulatory approval to compute market risk capital using their own VaR models rather than the standardised rulebook — typically producing 30–60% less capital for diversified books.
IRS	**Interest Rate Swap.** A derivatives contract in which two parties exchange a stream of fixed-rate payments for floating-rate payments (or vice versa) on a notional principal. The primary instrument used by ALM desks to manage duration gaps.
ISIN	**International Securities Identification Number.** A 12-character alphanumeric code that uniquely identifies securities globally.

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Abbreviation	Full Term and Brief Description
ISDA	**International Swaps and Derivatives Association.** The industry body that publishes standard legal documentation (Master Agreements, CSAs) and quantitative methodologies (SIMM) for over-the-counter derivatives markets.
KRI	**Key Risk Indicator.** A metric tracked continuously to signal that a model or process may be degrading. Examples: VaR backtesting exception rate, SIMM vs. actual IM reconciliation error.
KVA	**Capital Valuation Adjustment.** The present value of the cost of holding regulatory capital against a derivatives trade over its lifetime. Equal to the hurdle rate times the expected capital requirement, discounted to today.
LCH	**London Clearing House (LCH Group).** One of the world's largest central clearing counterparties, handling the majority of cleared interest rate swap volume globally.
LCR	**Liquidity Coverage Ratio.** A Basel III requirement: HQLA must cover at least 100% of net stressed cash outflows over a 30-day horizon. Ensures banks can survive a short-term liquidity crisis without central bank support.
LGD	**Loss Given Default.** The fraction of an exposure that is lost when a borrower defaults, after recovery from collateral and other sources. Typically 35–65% for unsecured corporate loans.
LMM	**LIBOR Market Model (also known as the BGM model).** A term-structure model that directly models the evolution of forward LIBOR/SOFR rates. Used for pricing callable derivatives and path-dependent instruments.
MBS	**Mortgage-Backed Securities.** Securities backed by pools of mortgage loans. Exhibit negative convexity due to borrower prepayment optionality, complicating ALM hedging.
MDD	**Model Development Document.** The SR 11-7 documentation artefact that captures a model's theoretical basis, implementation, validation results, limitations, use authorization, and open findings.
MVA	**Margin Valuation Adjustment.** The funding cost associated with posting initial margin over the life of a cleared or bilateral derivatives trade. Cleared trades generally attract lower MVA than bilateral.
NII	**Net Interest Income.** The difference between interest earned on assets and interest paid on liabilities. The primary revenue line for most commercial banks.
NIM	**Net Interest Margin.** NII expressed as a percentage of average earning assets. The headline profitability metric for comparing banks of different sizes. Typical range: 2–4% for US commercial banks.

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Abbreviation	Full Term and Brief Description
NLP	**Nonlinear Program (Nonlinear Programming).** An optimisation problem where the objective function or constraints are nonlinear. The full balance sheet optimisation is an NLP with stochastic and integer components.
NSFR	**Net Stable Funding Ratio.** A Basel III requirement: Available Stable Funding must exceed Required Stable Funding. Ensures that long-term assets are funded with long-term liabilities, not short-term borrowing.
OCI	**Other Comprehensive Income.** A component of equity that records unrealised gains and losses on AFS securities, derivatives designated as hedges, and other items not yet recognised in profit and loss.
OIS	**Overnight Indexed Swap.** An interest rate swap referencing an overnight rate (SOFR in the US, €STR in Europe). The OIS curve is the risk-free term structure used as the base for FTP rates and derivatives discounting.
OMS	**Order Management System.** The system that records, routes, and tracks trading orders and resulting positions. The real-time positions feed for the VaR and FRTB calculators.
OpRisk	**Operational Risk.** The risk of loss from inadequate or failed internal processes, people, systems, or external events (including fraud, fines, and cyber incidents).
PD	**Probability of Default.** The estimated likelihood that a borrower will fail to meet its debt obligations within a given horizon (typically one year). Combined with LGD to compute EL and with EAD for RWA.
PEP	**Politically Exposed Person.** An individual who holds or has held a prominent public function. Subject to enhanced AML due diligence under FATF standards.
PPNR	**Pre-Provision Net Revenue.** Revenue before loan loss provisions — NII plus non-interest income minus operating expenses. Used in DFAST projections to estimate the income buffer available to absorb stressed losses.
RAROC	**Risk-Adjusted Return on Capital.** Net income (after funding costs and expected losses) divided by economic capital. The primary metric for comparing profitability across business lines with different risk profiles.
ROA	**Return on Assets.** Net income divided by total assets. A high-level profitability metric; at JPMorgan scale, 5 basis points of ROA improvement is worth approximately $2 billion per year.
RORWA	**Return on Risk-Weighted Assets.** Net income divided by RWA. Preferred over ROA by bank management because it accounts for capital intensity — a business that earns the same income on lower RWA is more efficient.

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Abbreviation	Full Term and Brief Description
RSA	**Rate-Sensitive Assets.** Assets whose yield will reprice within a given period (typically 12 months) when interest rates change. Includes floating-rate loans, maturing fixed-rate instruments, and short-dated securities.
RSF	**Required Stable Funding.** The denominator of the NSFR. Assets weighted by their need for stable funding over a one-year horizon — long-term loans receive a high RSF factor; short-dated liquid assets receive a low one.
RSL	**Rate-Sensitive Liabilities.** Liabilities whose cost will reprice when interest rates change. Variable-rate deposits and short-term wholesale borrowings are highly rate-sensitive.
RWA	**Risk-Weighted Assets.** Assets multiplied by their regulatory risk weights. The denominator of all capital ratios. A measure of economic risk, not raw balance sheet size.
SA	**Standardised Approach.** The Basel regulatory method for computing capital charges using prescribed risk weights. Less capital-intensive to operate than IMA but typically produces higher capital requirements for diversified portfolios.
SIMM	**Standard Initial Margin Model.** The ISDA industry-standard model for computing initial margin on bilateral (uncleared) OTC derivatives. Version 2.6 covers six risk classes: IR, CRQ, CRNQ, EQ, FX, and Commodity.
SLR	**Supplementary Leverage Ratio.** A leverage ratio that uses total leverage exposure (assets plus off-balance-sheet items plus derivative add-ons) rather than RWA as the denominator. Catches banks that game risk weights.
SOFR	**Secured Overnight Financing Rate.** The US dollar risk-free rate based on overnight repo transactions secured by US Treasuries. Replaced LIBOR as the primary USD reference rate from June 2023.
SR 11-7	**Federal Reserve Supervisory Guidance on Model Risk Management (2011).** The foundational US regulatory guidance requiring banks to maintain rigorous model development, independent validation, and ongoing monitoring for all quantitative models used in material decisions.
SVaR	**Stressed Value at Risk.** VaR computed using a historical stress window (typically the 2007–2009 financial crisis). Added to regular VaR in the Basel 2.5 market risk capital formula to ensure capital does not fall too low in benign periods.
SVB	**Silicon Valley Bank.** A US regional bank that collapsed in March 2023, illustrating the consequences of duration mismatch (long HTM bonds funded by rate-sensitive deposits) and weak NSFR management.

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Abbreviation	Full Term and Brief Description
VaR	**Value at Risk.** The maximum expected loss on a portfolio over a given horizon at a given confidence level. At 99% over 1 day: there is a 1% chance of losing more than VaR on any given day.
VM	**Variation Margin.** Daily cash collateral exchanged between derivatives counterparties to reflect changes in the mark-to-market value of outstanding trades. Eliminates the accumulation of unrealised losses.
XVA	**X-Valuation Adjustment (collective term).** The family of valuation adjustments applied to derivatives to capture counterparty credit risk (CVA), own credit (DVA), funding (FVA), margin funding (MVA), capital (KVA), and collateral (ColVA).

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